Mobile Application Development

Practical No. 14

Design and develop any convertor application like temperature convertor /currency convertor/ volume convertor.

Que. 1 - Create a temperature converter from Celsius to Fahrenheit.

Que. 2. Extend the app to convert Fahrenheit to Celsius.

Que. 3. Design a currency converter with fixed exchange rate.

Que. 4. Implement a volume converter (liters ↔ gallons).

Que. 5. Add a spinner to select conversion type and update UI accordingly.

Que. 6. Validate input and show error messages for invalid entries.

ANS:-

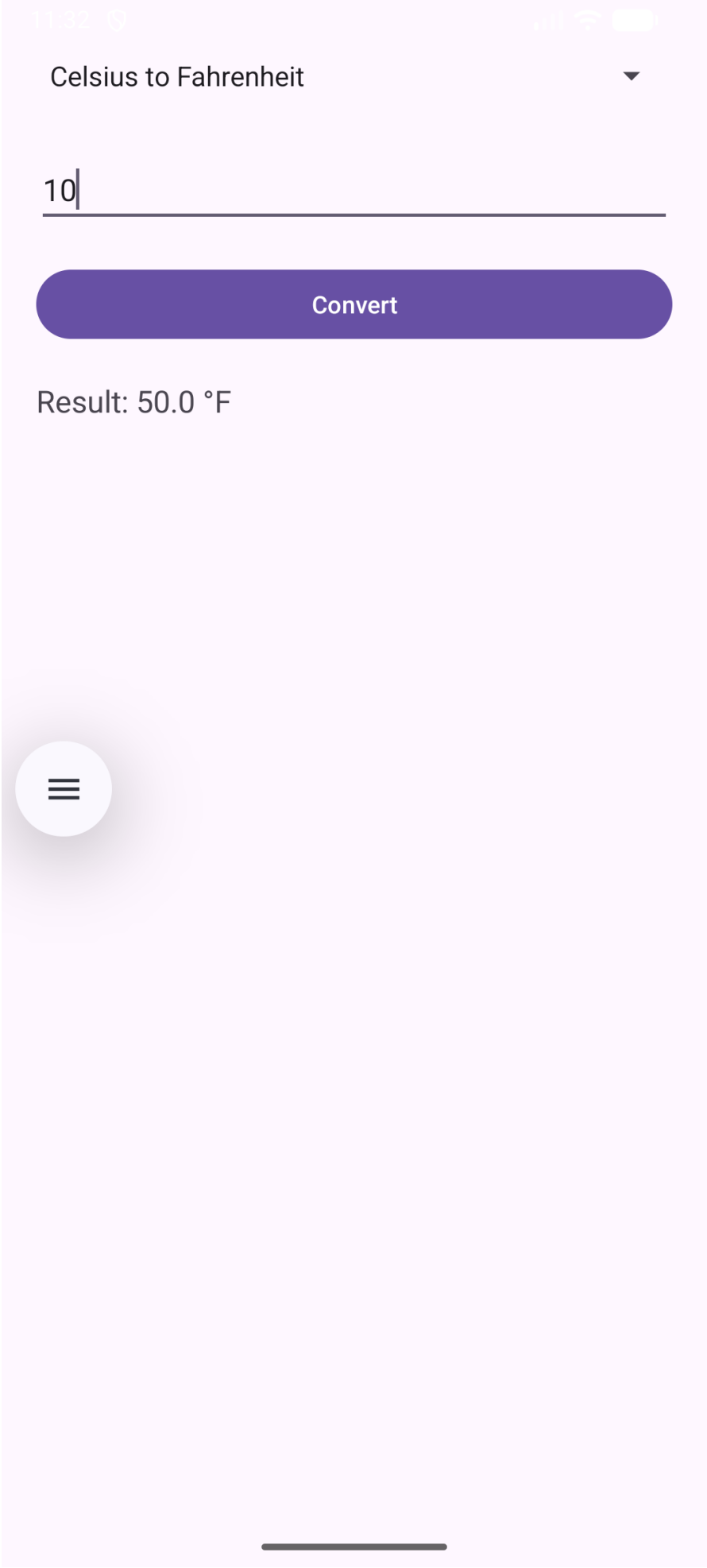
1 - activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <Spinner
        android:id="@+id/spinnerConversion"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter value"
        android:inputType="numberDecimal"
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/btnConvert"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Convert"
        android:layout_marginTop="20dp"/>

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Result"
```

```
        android:textSize="18sp"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

2 - MainActivity.java

```
package com.example.practical14;

import android.os.Bundle;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Spinner spinnerConversion;
    EditText etInput;
    Button btnConvert;
    TextView tvResult;

    // Fixed currency exchange rate (Example: 1 USD = 83 INR)
    final double USD_TO_INR = 83.0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        spinnerConversion = findViewById(R.id.spinnerConversion);
        etInput = findViewById(R.id.etInput);
        btnConvert = findViewById(R.id.btnConvert);
        tvResult = findViewById(R.id.tvResult);

        String[] conversionTypes = {
                "Celsius to Fahrenheit",
                "Fahrenheit to Celsius",
                "USD to INR",
                "INR to USD",
                "Liters to Gallons",
                "Gallons to Liters"
        };

        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_spinner_dropdown_item,
                conversionTypes
        );

        spinnerConversion.setAdapter(adapter);

        btnConvert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                String inputStr = etInput.getText().toString().trim();

                if (inputStr.isEmpty()) {
                    etInput.setError("Please enter a value");
                    return;
                }

                double inputValue;

                try {
                    inputValue = Double.parseDouble(inputStr);
                } catch (NumberFormatException e) {
```

```
                    etInput.setError("Invalid number");
                    return;
                }

                String selected =
spinnerConversion.getSelectedItem().toString();
                double result = 0;

                switch (selected) {

                    case "Celsius to Fahrenheit":
                        result = (inputValue * 9/5) + 32;
                        tvResult.setText("Result: " + result + " °F");
                        break;

                    case "Fahrenheit to Celsius":
                        result = (inputValue - 32) * 5/9;
                        tvResult.setText("Result: " + result + " °C");
                        break;

                    case "USD to INR":
                        result = inputValue * USD_TO_INR;
                        tvResult.setText("Result: ₹ " + result);
                        break;

                    case "INR to USD":
                        result = inputValue / USD_TO_INR;
                        tvResult.setText("Result: $ " + result);
                        break;

                    case "Liters to Gallons":
                        result = inputValue * 0.264172;
                        tvResult.setText("Result: " + result + " Gallons");
                        break;

                    case "Gallons to Liters":
                        result = inputValue / 0.264172;
                        tvResult.setText("Result: " + result + " Liters");
                        break;
                }
            }
        });
    }
}
```

Output:-